

QALA

Universal Solution Management OS

UI/UX Design System · 22 Screens · Dark Mode · Multi-Domain

22 Screens

Dark Mode

6 Domains

12 User Flows

Document Type: Screen Flows & Annotated Wireframes · Version 1.0
System: Qala Universal Solution Management OS
Companion Docs: SDD v2 · LLD v1 · RFC v2 · PRD · UI/UX Spec v1

1. Introduction & Design System

This document presents 22 high-fidelity screen designs for the Qala Universal Solution Management OS. Each screen is rendered in dark mode using the Qala design system and is accompanied by a full annotation table describing every interactive element, state, and accessibility consideration.

Element	Description	State / Behaviour
Background	#0B0F19 (BG) · #111827 (BG2) · #16213A (BG3)	Three-tier dark surface hierarchy
Accent / Action	#388BFD Action Blue	All interactive affordances: buttons, links, focus rings
Success	#22C564 Green	Gate passed, approved, active, deployed states
Warning	#F59E0B Amber	Medium risk, awaiting review, pending action
Danger	#EF4444 Red	Error, critical alert, quarantined, recall
AI Feature	#8B5CF6 Purple	AI Agent indicator, recommendation cards, AI actions
Domain / Teal	#14B8A6 Teal	Active state, domain extension, workbench type
Text Primary	#F0F5FF Near-white	All primary body text, headings on dark surfaces
Text Secondary	#CBD5E1 Slate 300	Metadata, captions, placeholder, hint text
Typography	DejaVu Sans + DejaVu Sans Mono	Sans for all UI; Mono for IDs, versions, checksums, logs
Border Radius	4px (inputs) · 8px (buttons/cards) · 10px (major cards) · 14px (modals)	Progressive radius scale
Spacing Base	8px grid	All spacing in multiples of 8px

Persona-Adaptive Vocabulary Layer

Qala serves 8 domain personas on a single unified platform. The vocabulary layer activates at login and adapts sidebar labels, empty-state messaging, and page titles based on the user's domain. The underlying component structure is identical across all domains.

Element	Description	State / Behaviour
Platform Term	Software	CPG / Pharma
Development Environment	SDE	Formulation Workbench
Build / Compile	Build	Simulation Run
Quality Gate	CI Quality Gate	Batch Release Gate
Release	Software Release	Batch Release
Artefact	Container Image / Package	Specification / CoA

2. User Flow Map

The following table maps each of the 12 primary user flows to the screens they traverse. Screen identifiers link to the annotated screen designs in Section 3.

Element	Description	State / Behaviour
UF-01 New User Onboarding	SC-01 Login → SC-02 Onboarding → SC-03 Dashboard → SC-06 SDE Detail → SC-07 Pipeline	Onboarding
UF-02 CPG Change Control	SC-04 Solutions → SC-08 CCR Detail → SC-06 Workbench → SC-09 Release	CPG Governance
UF-03 Software Release	SC-05 Solution → SC-07 Pipeline → SC-09 Release Gates → SC-09 Deploy	Software Release
UF-05 Security Incident	SC-19 Quarantine → SC-19 Forensics → SC-06 Recovery → SC-10 Audit	Security Response
UF-06 AI Recommendation	SC-13 AI Panel → SC-08 CCR Auto Applied → SC-03 Dashboard	AI-Assisted
UF-12 Compliance Exam	SC-11 Evidence → SC-11 Regulator → SC-10 Audit → SC-21 Notify	Compliance

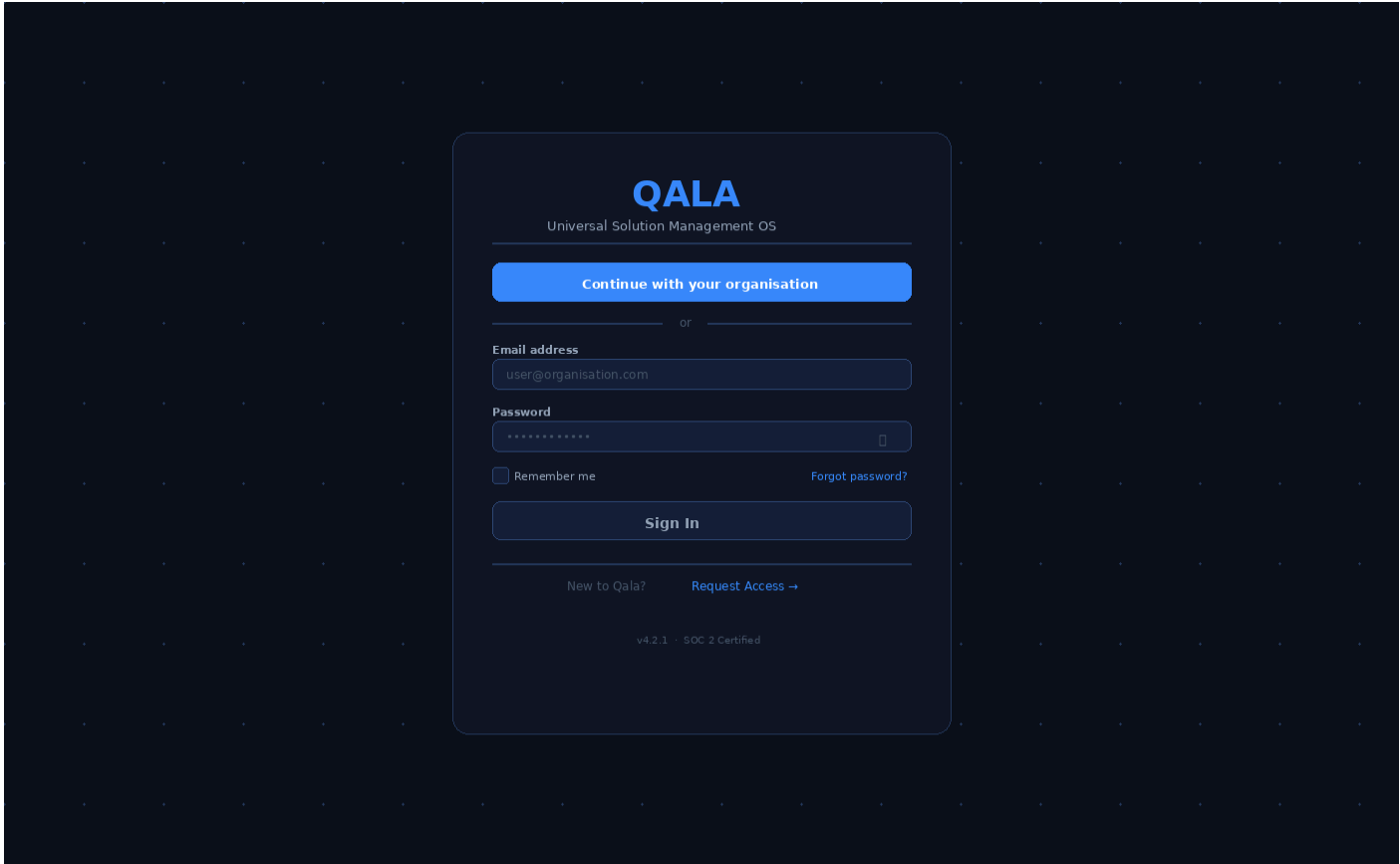
3. Annotated Screen Designs

SC-01

Login Screen

UF-01 New User Onboarding → Unauthenticated Entry Point

All Domains



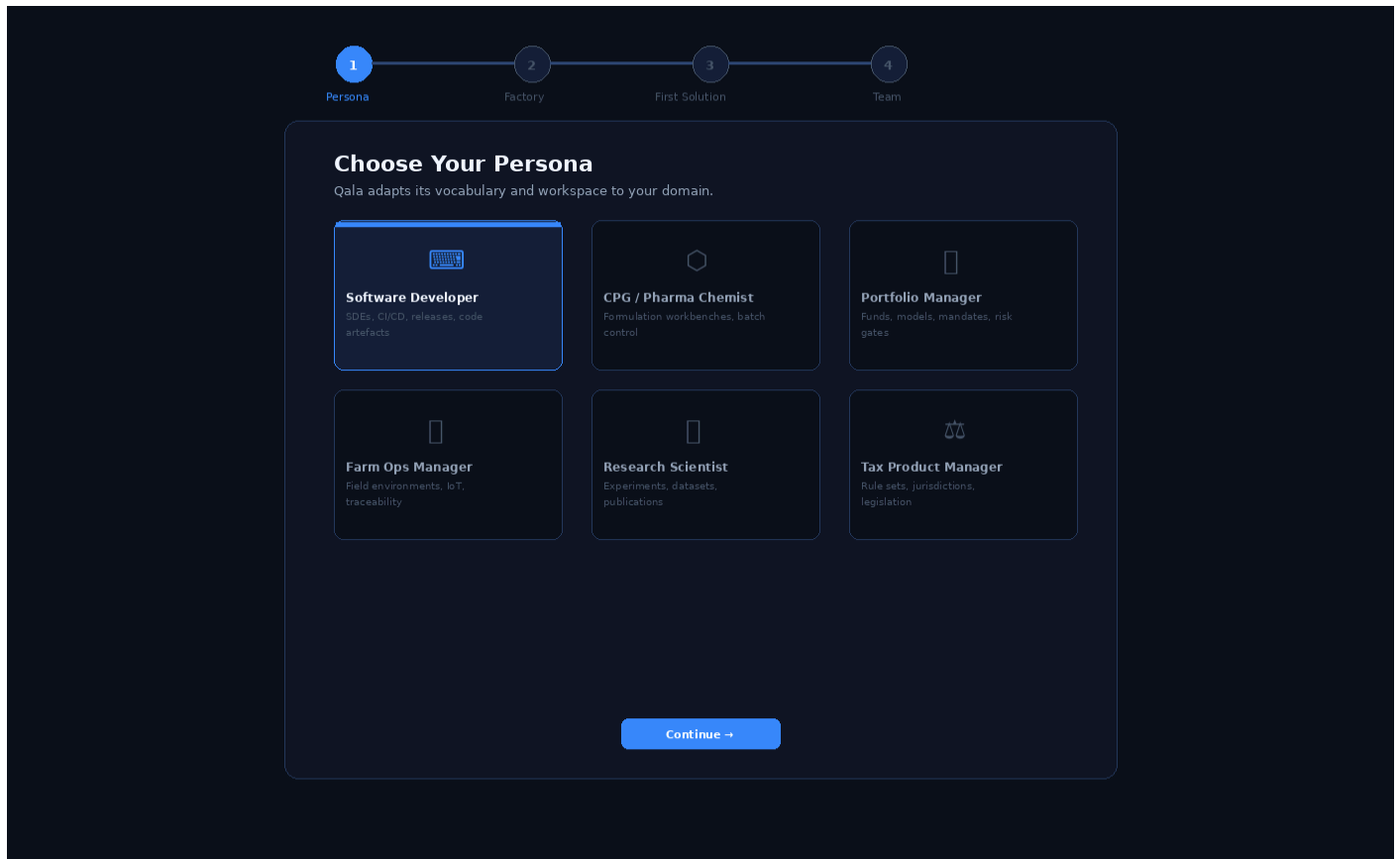
Design Notes

Full-bleed Brand Navy background (#0B0F19). Card uses elevated surface with 16px radius. Subtle dot-grid background texture creates depth without distraction. WCAG AA on all text/button combinations.

Element Annotations

Element	Description	State / Behaviour
Logo + Tagline	QALA wordmark centred on Brand Navy card (480px wide)	Static — not a link in unauthenticated state
SSO Button	Primary full-width button — triggers SAML/OIDC redirect	Autofocus on load; org domain pre-filled if detectable
Email Input	Standard email type input	Validates on blur; autofocus if SSO not detected
Password Input	type=password with eye-toggle visibility	Inline error on invalid credentials below field

Element	Description	State / Behaviour
Remember Me	Checkbox + Forgot Password link on right	Forgot opens email-entry modal (separate flow)
Sign In Button	Secondary primary button	Disabled until both fields non-empty; spinner on submit
Request Access	Link to public marketing trial form	Routes outside the authenticated app



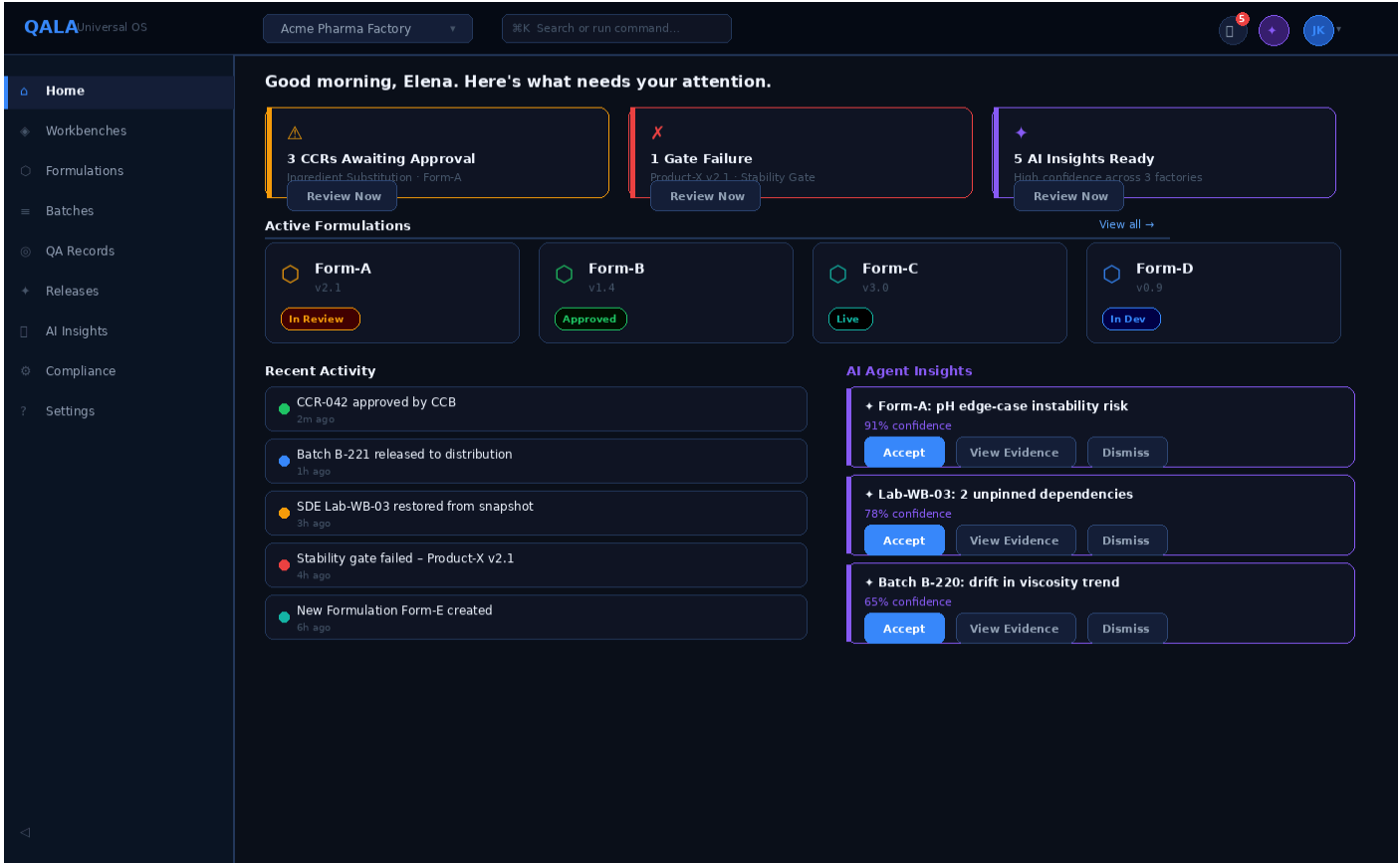
Design Notes

Wizard uses a centred card layout (840px) on the dark canvas. Persona cards use icon + name + one-line description pattern. The selected card has a 2px blue border + top accent bar to show selection clearly without colour alone.

Element Annotations

Element	Description	State / Behaviour
Progress Steps	4-step horizontal progress indicator	Step 1 active (filled blue); remaining steps grey
Persona Grid	6 persona cards in 3-column grid	Selected card gains blue border + accent rule; card describes domain vocabulary
Software Developer	SDEs, CI/CD, releases, code artefacts	Default selection for new users without org context
CPG / Pharma Chemist	Formulation Workbenches, Batch Control	Vocabulary swaps to CPG terms on selection
Portfolio Manager	Funds, Models, Mandates, Risk Gates	Financial domain — backtests replace builds

Element	Description	State / Behaviour
Farm Ops Manager	Field Environments, IoT, Traceability	Mobile-first layout preferred for this persona
Continue Button	Primary CTA — advances to Step 2 (Factory setup)	Enabled immediately; no selection required to proceed



Design Notes

Dashboard uses a 3-priority-card + 4-solution-card + 2-column (feed + AI) layout. The sidebar vocabulary reads "Workbenches", "Batches" — CPG domain adaptations. Priority cards use left-border colour coding with BG2 card fill.

Element Annotations

Element	Description	State / Behaviour
GlobalNav	64px persistent top bar with factory switcher, search, notifications, AI orb, avatar	Scrolled state adds elevation shadow; notification badge shows unread count
Priority Cards	3 cards showing action-required items with colour-coded severity borders	Amber = action, Red = blocking, Purple = AI feature
CCR Approval Card	Count of CCRs awaiting the logged-in user's approval	Click routes to CCR list pre-filtered to "Awaiting My Approval"
Gate Failure Card	Solution name + version, click to release gate detail	Red accent = release blocked; immediate action required
AI Insights Card	Count of pending AI recommendations	Purple accent; opens AI Agent insights panel drawer

Element	Description	State / Behaviour
Active Formulations	Last 4 accessed solutions as compact cards — vocabulary adapted per persona	CPG persona sees "Formulations" not "Solutions"
Event Feed	Last 20 events across all factories, relative timestamps	Slide-in animation for new events; "Load more" paginates
AI Recommendation Panel	Top 3 AI recommendations with inline Accept/Dismiss	Accept triggers confirmation then auto-creates CCR

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247 solutions

+ New Formulation

Filter

Sort: Updated

Search formulations...

Name	Type	State	Version	Owner	Updated	
○ Formulation-A	Liquid Emulsion	In Review	v2.1.0	Elena V.	2m ago	...
○ Formulation-B	Cream Base	Approved	v1.4.2	Marco D.	1h ago	...
○ Formulation-C	Serum	Live	v3.0.0	Sara T.	3h ago	...
○ Formulation-D	Gel Formula	In Dev	v0.9.1	Priya S.	6h ago	...
○ Formulation-E	Suspension	Draft	v0.1.0	Elena V.	1d ago	...
○ Formulation-F	Emulsion	Released	v4.2.0	Marco D.	2d ago	...
○ Formulation-G	Powder Base	Deprecated	v1.0.0	Sara T.	5d ago	...
○ Formulation-H	Lotion	Recalled	v2.0.0	Admin	1w ago	...

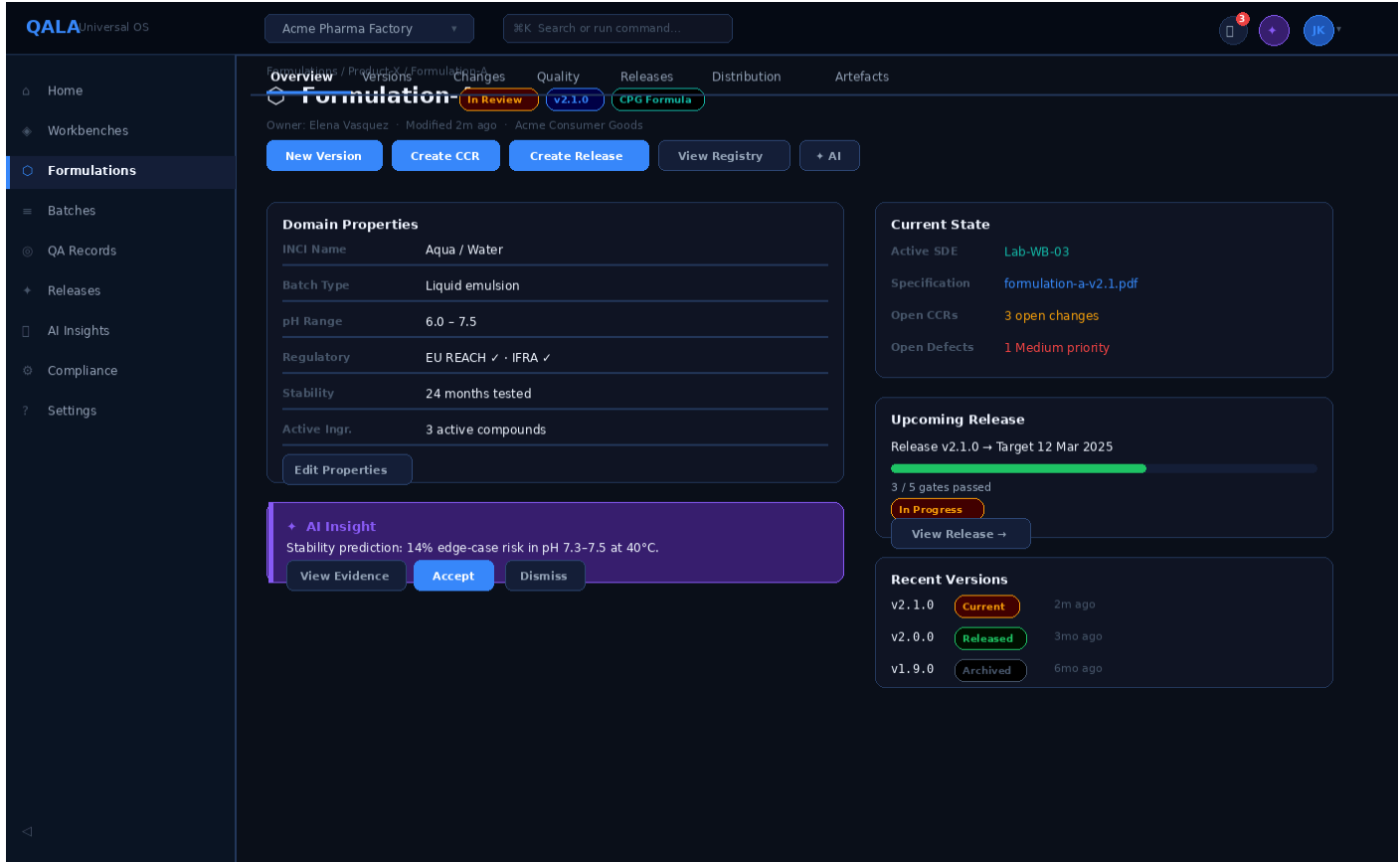
Design Notes

Table uses alternating BG2/BG row fill. Active row gets 1px blue outline. The breadcrumb trail and page title use H2 styling. Chip labels never rely on colour alone — all states have distinct text labels.

Element Annotations

Element	Description	State / Behaviour
Toolbar	New Formulation (primary), Filter, Sort controls + search bar	Filter expands to multi-select panel; sort persists per user
Table Header	Sticky header: Name, Type, State, Version, Owner, Updated, Actions	Sortable by clicking column header; active sort shows chevron
Lifecycle State Chips	Colour-coded chips: In Review (amber), Approved (green), Live (teal), Draft (grey)	Click chip opens State Transition drawer directly from list
Version Column	Semver in monospace font	Click opens version history drawer
Owner Column	Name + avatar — links to user profile	Hover shows full name and last-active status

Element	Description	State / Behaviour
Actions Column	... kebab opens: View, Edit, Clone, Create CCR, Archive	Archive requires confirmation dialog
Selected Row Highlight	First row highlighted with blue outline	Multi-select via Space key; batch actions appear in toolbar



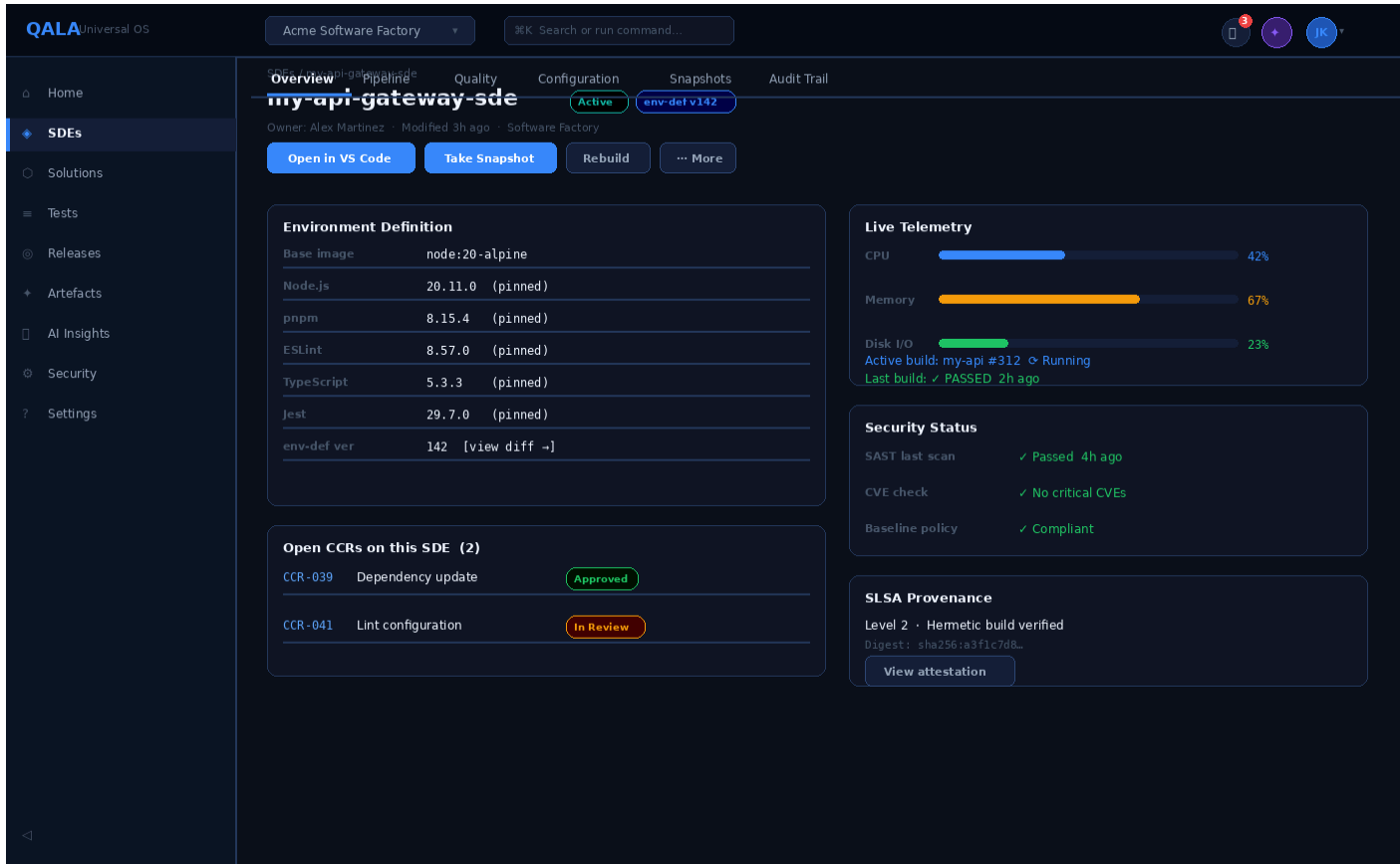
Design Notes

Two-column layout: 580px left (domain properties + AI insight) and 460px right (state + release + versions). All panels use BG2 card fill with BORDER outline. The AI insight panel uses PURPLE_DIM fill with left accent border.

Element Annotations

Element	Description	State / Behaviour
Header + State Chip	Solution name H2 + lifecycle state chip + version badge	State chip click opens State Transition drawer (SC-20)
Action Bar	New Version (primary), Create CCR, Create Release, View Registry, AI contextual actions	Primary action adapts to solution type and current state
Tab Bar	Overview, Versions, Changes, Quality, Releases, Distribution, Artefacts	Active tab shown by underline in Action Blue; keyboard arrow navigation
Domain Properties Panel	Fields rendered from CPG Domain Pack schema (INCI Name, pH, batch type, etc.)	Edit mode opens inline; schema-driven — no code changes for new domains

Element	Description	State / Behaviour
Current State Panel	Active SDE link, current spec artefact, open CCR count, open defect count	Each item is a navigable link to the referenced record
Upcoming Release Card	Release version, target date, gate progress bar, status chip	Progress bar fills proportionally to gates passed
AI Insight Banner	Purple-accented card with AI recommendation for this specific solution	View Evidence expands full analysis; Accept auto-creates CCR
Versions Mini-panel	Latest 3 versions with semver, state chip, timestamp	Click any version to navigate to that version's detail



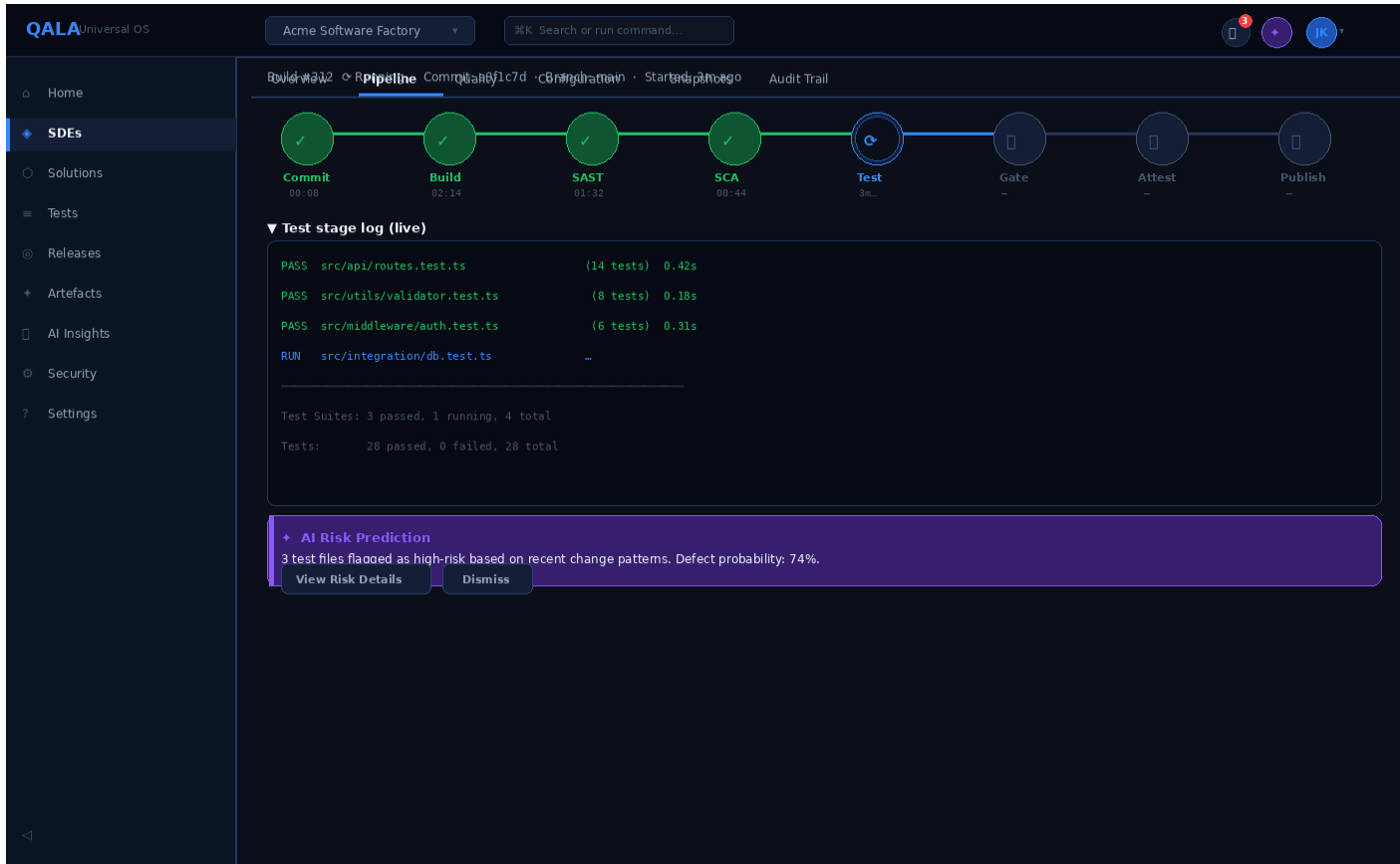
Design Notes

Software SDE shows live telemetry and pipeline sections not present in other domain SDE types. The domain-adaptive sidebar shows "SDEs" label for software persona. Environment Definition uses monospace font for all version values.

Element Annotations

Element	Description	State / Behaviour
SDE Header	SDE name as H2, Active chip (teal), env-def version badge	Tab "SDE" shows as "SDE" for software; "Formulation Workbench" for CPG
Primary Action	"Open in VS Code" launches IDE with Qala Dev Extension	Replaced by "Open Simulation" for CPG; "Open Jupyter" for Financial
Environment Definition	All tool versions listed with pinned/unpinned status	"view diff" link shows changes from prior env-def version
Live Telemetry	CPU, memory, disk I/O bars updating every 5 seconds	Only visible for software SDEs; replaced by LIMS data for CPG
Security Status	SAST, CVE, baseline policy results with timestamps	Red X on any item opens findings report in right drawer

Element	Description	State / Behaviour
SLSA Provenance	Build level + hermetic verification + SHA-256 digest	Click "View attestation" opens full provenance chain
Open CCRs Panel	2 CCRs linked to this SDE with state chips	Click any CCR row opens CCR Detail in a right drawer



Design Notes

Pipeline uses a horizontal node-connector layout. The stage nodes use 52px circles. The log panel uses a near-black (#060A14) background with monospace font for terminal output. The AI risk banner uses PURPLE_DIM fill with left accent border.

Element Annotations

Element	Description	State / Behaviour
Build Metadata Strip	Build number, status badge, commit SHA, branch, elapsed time	Live updates every 2 seconds; SHA links to source commit
Stage Nodes	8 connected pipeline stages: Commit → Build → SAST → SCA → Test → Gate → Attest → Publish	Green = Passed, Blue animated ring = Running, Grey = Pending, Red = Failed
Running Stage	Blue animated border ring on active stage node	Reduced motion: static blue border replaces animation
Stage Log Panel	Live streaming log output for the active stage	Copy button, filter input, auto-scroll toggle

Element	Description	State / Behaviour
Stage Connector Lines	Horizontal lines between nodes coloured by upstream stage result	Turns green as stages pass; red on failure stops progression
AI Risk Overlay	Purple banner: "3 test files flagged as high-risk — 74% defect probability"	Based on historical change patterns for affected files; dismissable
Duration Labels	Per-stage elapsed time shown below nodes	Shows "—" for pending stages; "3m..." for running stages

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Change Control / CCR-042: Ingredient Substitution

CCR-042

Ingredient Substitution

Risk Score: 8 / 10 HIGH

Awaiting CCB

Stage 2 of 3

Submitted by: Elena Vasquez · 2025-03-07 · 3 factories affected

Approve

Request Changes

Reject

Escalate to CCB

Change Description

Substitute Palm Oil Glycerides with Shea Butter

Glycerides in Products: Form-A, Form-B, Form-C.

Justification:
Global palm oil supply chain disruption. Certified sustainable alternative with equivalent performance validated in Sim-031.

Implementation Plan:
1. Update formulation spec in SDE
2. Run stability simulation x3
3. Upload simulation artefacts
4. Regulatory pre-check EU REACH

Regulatory Implications

EU REACH: Re-notification required within 30 days

US FDA 21 CFR: No change required

IFRA: Compliance check in progress...

Approval Chain

Stage 1: Formulation Chemist
Elena V. · Approved 09:14

Stage 2: QA Manager ← current
Marco D. · Pending approval

Stage 3: Change Control Board
Pending Stage 2 completion

Impact Analysis

14 affected solutions across 3 product lines

Form-A HIGH View →

Form-B MEDIUM View →

Form-C MEDIUM View →

Form-D LOW View →

+10 more

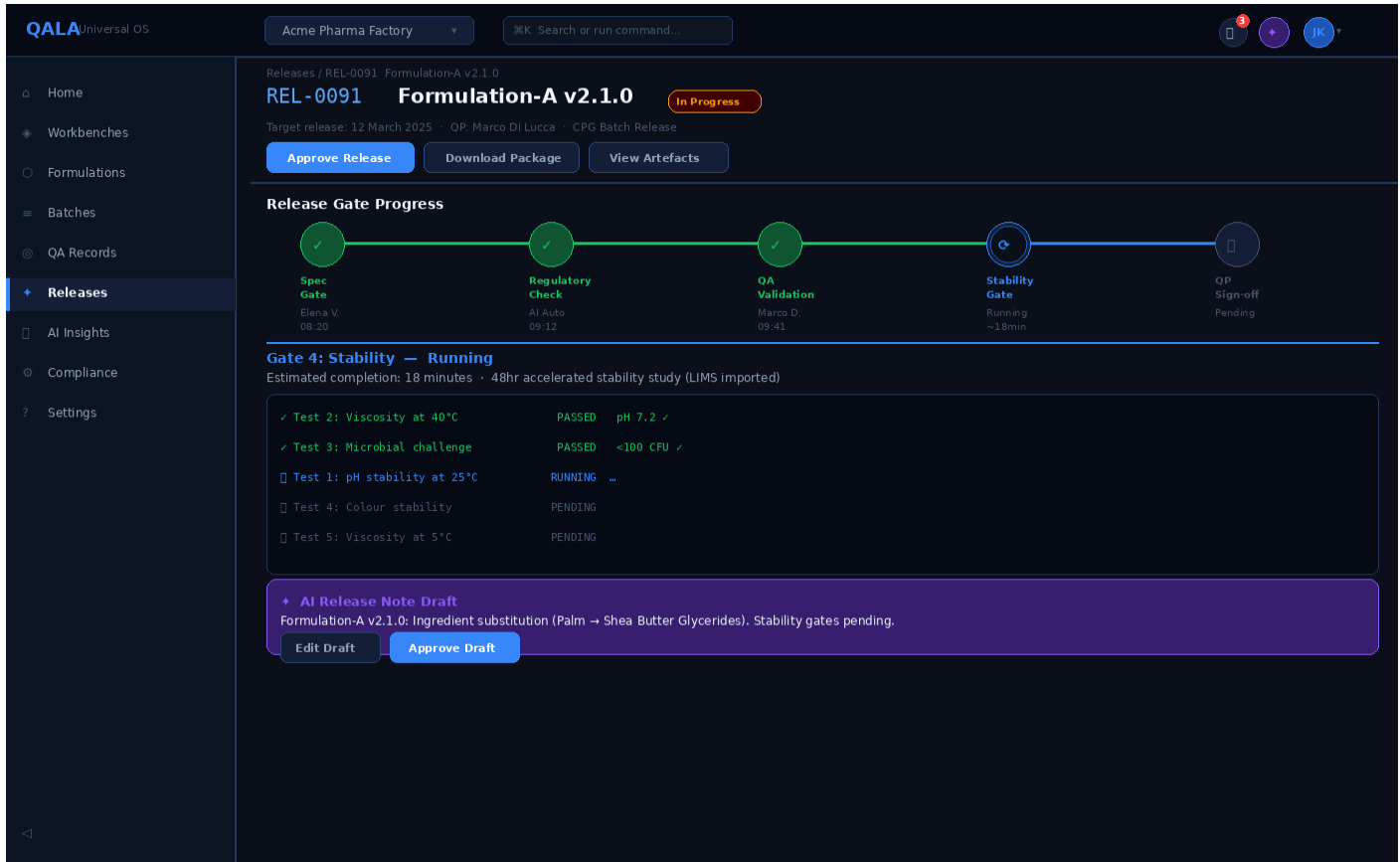
Design Notes

CCR Detail uses a two-column layout. The Risk Score badge uses RED_DIM fill with RED border for HIGH risk. The Approval Chain is a vertical timeline with connecting lines between stage circles. All action buttons adapt to the logged-in user's role in the approval chain.

Element Annotations

Element	Description	State / Behaviour
CCR ID + Title	Monospace CCR identifier + plain-text title as H2	ID links to CCR history; title editable by requester only in Draft state
Risk Score Badge	"8 / 10 HIGH" badge — computed from change type, impact scope, regulatory flags	Score breakdown panel opens on click/hover; colour transitions animated
Approval Chain	Vertical timeline: Stage 1 (done), Stage 2 (current), Stage 3 (pending)	Current stage highlighted; avatar + name + timestamp per stage
Action Bar	Approve, Request Changes, Reject, Escalate — shown only to current stage approver	Other users see read-only action set (View Evidence, Download)
Change Description	Rich text (read-only in approval view) + implementation plan numbered steps	Implementation plan editable by requester only while in Draft

Element	Description	State / Behaviour
Impact Analysis Panel	14 affected solutions listed with HIGH / MEDIUM / LOW severity chips	"View Relationship Graph" opens interactive dependency graph in new panel
Regulatory Implications	Domain-Pack-driven section — shows only when solution type has regulatory dimension	EU REACH re-notification in amber; items requiring no action in green



Design Notes

The gate strip uses the same node-connector pattern as the pipeline (SC-07). The running gate detail panel appears below the strip and shows the live test log in the same near-black terminal style. The AI release notes banner uses PURPLE_DIM with a distinct "Draft" label.

Element Annotations

Element	Description	State / Behaviour
Release Header	Release ID (monospace), solution + version, target date, QP name	Breadcrumb shows path to parent solution
Gate Progress Strip	5 gate nodes: Spec, Regulatory, QA Validation, Stability, QP Sign-off	Gates adapt to solution type — CPG has Stability + QP; software has SAST + Coverage
Passed Gates	Green filled circle + ✓ + evaluator name + timestamp + artefact link	Click gate node for full evidence details in drawer
Running Gate	Blue animated ring + "Running" + estimated completion time	Live test log streams below the strip while gate is active
Pending Gates	Grey circle + label — cannot be activated until upstream gates pass	Cannot be manually overridden without elevated Emergency CCR

Element	Description	State / Behaviour
Live Test Log	Terminal-style log showing test IDs, status, and result values	Auto-scrolls to latest output; can be paused; copy button available
AI Release Notes	AI auto-drafts release notes from linked CCRs and test summaries	Edit Draft opens rich-text editor; Approve Draft locks the content

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Audit Trail

Filter by solution or ID...Event Type▼Actor▼Date Range▼Export CSVExport JSONCompliance Report

3,847 events · Showing: last 30 days · Filtered: all event types

Timestamp	Event Type	Resource	Actor	Summary
2025-03-07 09:14:22	CCR_APPROVED	CCR-042	Elena Vasquez	Approved at stage 1
2025-03-07 09:02:15	ARTEFACT_PUBLISHED	Form-A-spec-v2.1	System	Spec published to registry
2025-03-07 08:57:43	CCR_SUBMITTED	CCR-042	Marco Di Lucca	Submitted ingredient change
2025-03-07 08:32:11	SDE_SNAPSHOT	Lab-WB-03	Elena Vasquez	Manual snapshot taken
2025-03-07 07:55:29	BUILD_PASSED	api-gateway #312	CI Agent	All 28 tests passed
2025-03-07 07:22:01	SECURITY_ALERT	api-gateway-sde	SEM	Anomalous egress detected
2025-03-07 06:44:18	AI_RECOMMENDATION	Form-A	AI Agent	pH instability risk flagged
2025-03-06 23:10:55	CCR_APPROVED	CCR-041	Tech Lead	Lint config update approved

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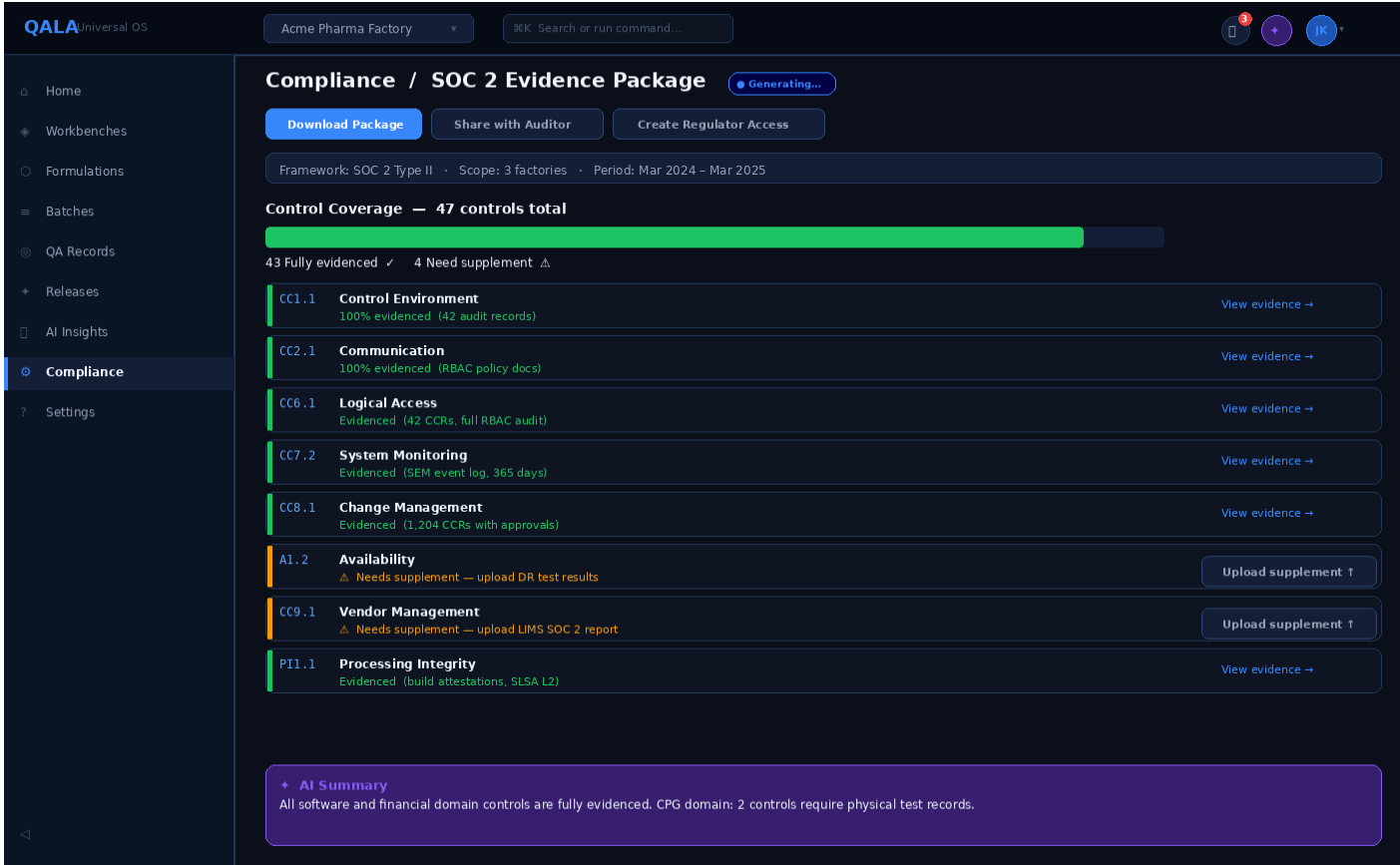
Design Notes

The audit trail table uses FONT_MONO for timestamps and event type identifiers to ensure clear reading of system-generated identifiers. The expanded row payload panel uses a darker (060A14) background to visually separate it from the main row.

Element Annotations

Element	Description	State / Behaviour
Filter Bar	Real-time filter: solution ID input, event type multi-select, actor autocomplete, date range picker	All filters are additive AND; active filters shown as chips with × dismiss
Export Actions	CSV, JSON exports honour current filter state; Compliance Report generates formatted evidence PDF	CSV includes all event payload fields; JSON is the raw canonical event envelope
Result Summary	"3,847 events · Showing: last 30 days · Filtered: all event types"	Summary updates in real time as filters change
Event Row	ISO 8601 timestamp · event type badge (monospace) · resource link · actor · summary	Alternating BG2/BG row fill; first row expanded showing full payload

Element	Description	State / Behaviour
Expanded Payload	Event ID, facility region, masked IP, full contextual data	Expand toggle is chevron on left of each row; keyboard accessible
Event Type Badge	Colour-coded chip: CCR_APPROVED (green), BUILD_PASSED (blue), SECURITY_ALERT (red), AI_RECOMMENDATION (purple)	Badge text is monospace; colour + text never colour alone
Pagination	Page-based for large audit trails; 25 per page default; virtual scroll on lg+ screens	Direct page navigation input available; prev/next keyboard-accessible



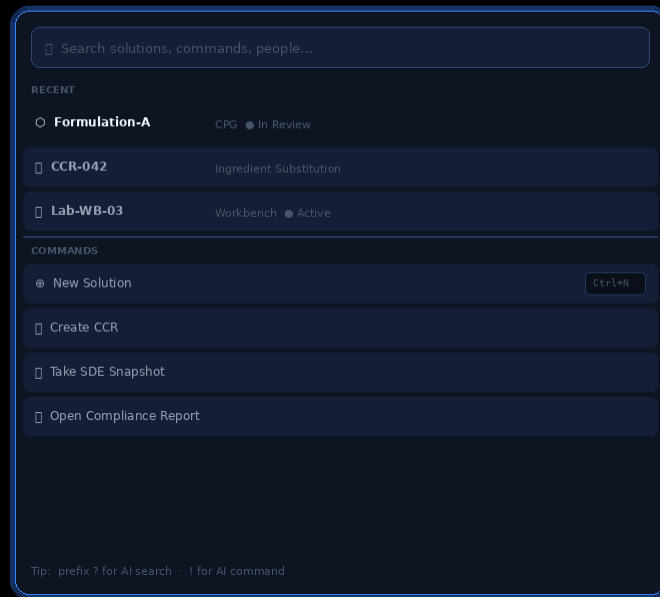
Design Notes

Each control row has a 4px left accent border (green = evidenced, amber = needs supplement). The coverage bar is 20px height to give visual weight to the overall posture metric. The AI summary uses the same PURPLE_DIM + PURPLE border pattern as all AI features.

Element Annotations

Element	Description	State / Behaviour
Header + Status	Evidence Package title + generating status chip + scope bar	Status changes from "● Generating..." to "● Complete" when done; estimated time shown
Download / Share / Regulator Access	Three primary actions on the package	Regulator Access creates a time-limited read-only portal with a secure one-time link
Coverage Progress Bar	Progress bar showing 43/47 controls evidenced (91%)	Green fill; 4 amber controls shown separately below

Element	Description	State / Behaviour
Evidenced Controls	Green tick + control ID (monospace) + control name + evidence summary	"View evidence" opens a detail drawer showing specific records (timestamps, actor names, CCR IDs)
Supplement Required	Amber warning + specific missing item + upload slot	Platform identifies exactly what is missing; upload slot appears inline for drag-and-drop
Control Row	Left accent border colour indicates status; click row for detail	All evidence records are linked to immutable audit events
AI Summary	Purple banner synthesising coverage state in plain English	Identifies which domains have gaps; AI cannot modify evidence records



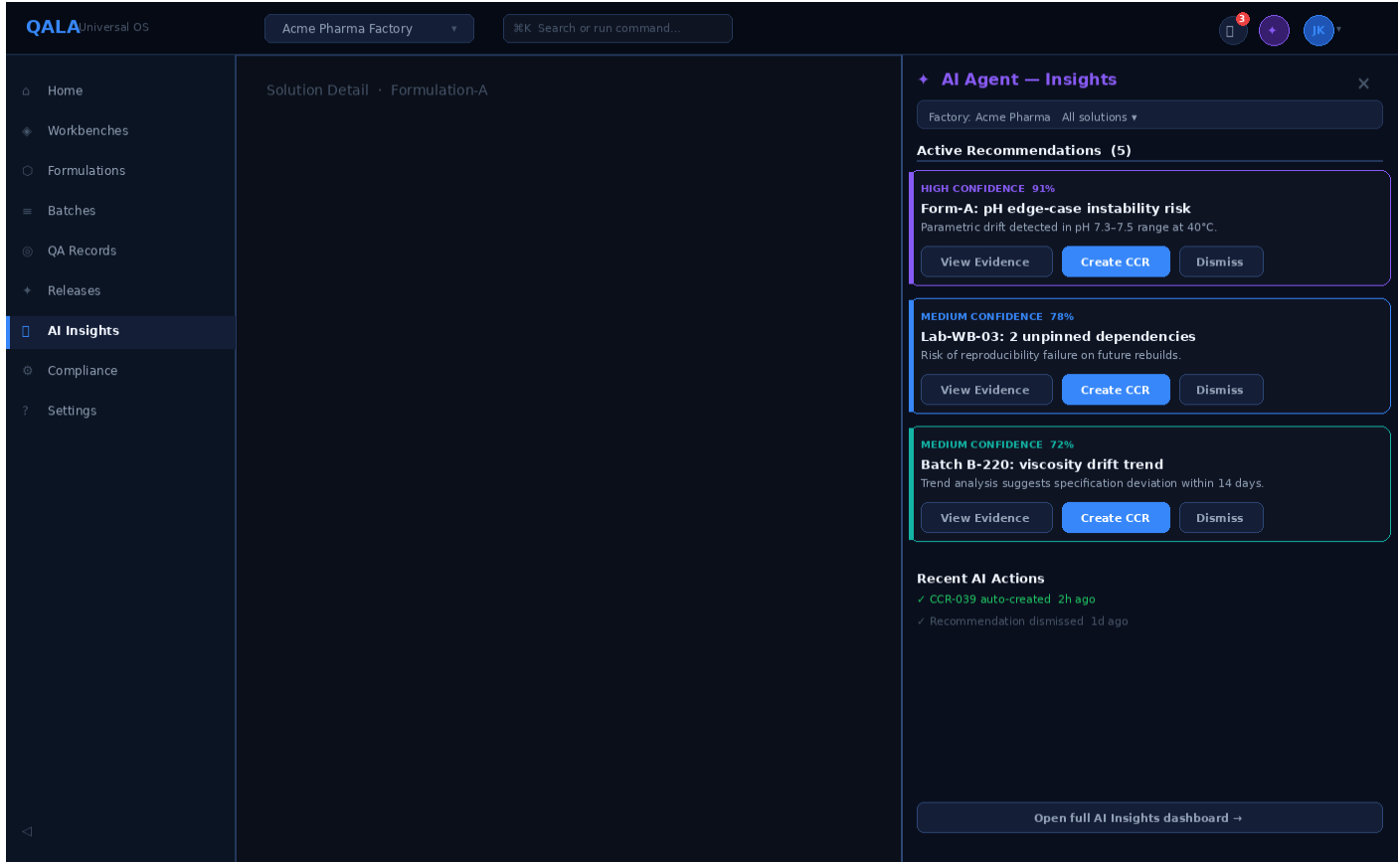
Design Notes

Command palette uses a 600px × max-600px container centred on screen with elevation.4 shadow and backdrop blur. Results have a minimum height of 38px per item. The blue glow border effect reinforces the "active overlay" state. Typography is slightly larger (13px) than standard body to improve scanability at speed.

Element Annotations

Element	Description	State / Behaviour
Overlay Background	Blurred + darkened background — backdrop filter blur(8px) rgba(0,0,0,0.4)	Clicking outside dismisses palette; Esc also dismisses
Search Input	Autofocus on open, debounced 50ms, full-width within palette (600px wide)	Placeholder adapts to persona; results group-formatted below
Recent Group	Last 8 accessed records across solutions, CCRs, workbenches	Arrow-key navigable; Enter opens selected record
Commands Group	Platform commands with keyboard shortcuts	Domain-specific commands load per active persona context
People Group	Team members with role labels — click to open profile or send message	Search "people:" prefix filters to team members only

Element	Description	State / Behaviour
AI Search Prefix	"?" prefix enables semantic AI-powered search across all content	"!" prefix sends a command directly to AI Agent (e.g. "!What is the risk on CCR-042")
Result Grouping	Groups: Recent · Solutions · CCRs · SDEs · Commands · People · AI	Groups collapse/expand; keyboard navigates between groups with Tab



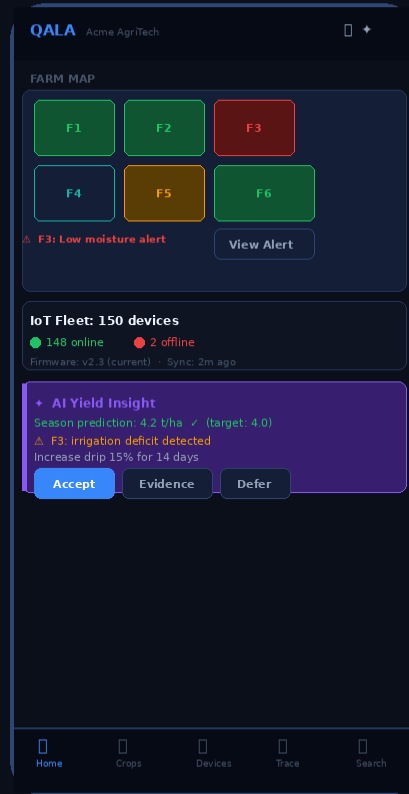
Design Notes

The AI panel is a 500px right-side drawer that appears on top of the current page content. All AI-themed elements use the PURPLE / PURPLE_DIM colour token pair. The panel header uses a slightly lighter background than the main drawer to establish visual hierarchy.

Element Annotations

Element	Description	State / Behaviour
AI Orb Button	Purple orb in GlobalNav triggers the right-side AI panel drawer	Orb animates when recommendations are available; static when no new insights
Scope Selector	Factory + solution type filter at the top of the panel	"All solutions" shows platform-wide; narrow to single solution for focused view
HIGH CONFIDENCE Card	Top recommendation with 91% confidence bar, evidence summary, 3 action buttons	Confidence bar uses PURPLE fill; HIGH = purple accent, MEDIUM = blue accent
Accept / Create CCR	Accept button pre-populates CCR form with recommendation data	Low-risk recommendations auto-approve; high-risk route to approval chain

Element	Description	State / Behaviour
Dismiss Action	Dismisses card with fade animation; stored in AI feedback log	Dismiss reason can be added (optional); improves AI model over time
Recent AI Actions	Shows platform actions taken by AI Agent (auto-created CCRs, dismissed recommendations)	Creates trust by making AI activity transparent and auditable
Full Dashboard Link	Routes to the AI Insights full-page view with trend charts	Full dashboard shows confidence history, domain coverage, and feedback stats



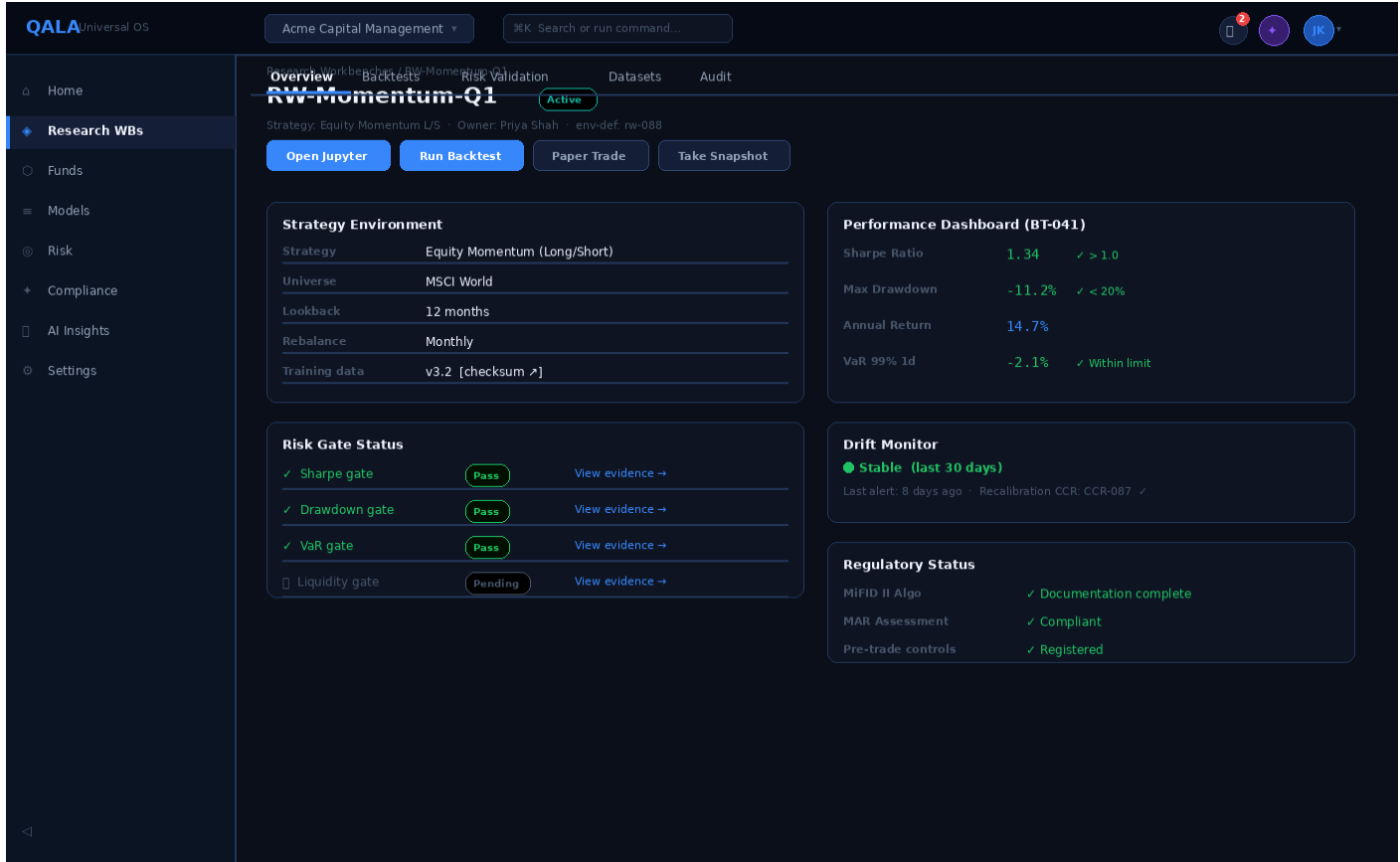
Design Notes

The farm dashboard uses a mobile-portrait frame (400×780px) rendered within the landscape canvas. Bottom tab navigation replaces the sidebar for mobile breakpoints. The farm map uses filled polygon shapes with colour borders — red/amber/green use high-contrast pairs with distinct labels, never colour alone.

Element Annotations

Element	Description	State / Behaviour
Mobile Nav Bar	Compact top bar: Logo, factory name, notification + AI icons	Hamburger ≡ opens full sidebar drawer; factory name truncated on small screens
Farm Map	Field section polygons colour-coded by moisture / health status	Green = normal, Amber = caution, Red = alert; tap section opens detail drawer
Field Alert	Red alert below map: "F3: Low moisture alert" with View Alert button	Alert triggered by IoT sensor threshold; links to sensor detail and irrigation controls
IoT Fleet Status	Total device count, online/offline split, firmware version, last sync	Offline count shown in red; firmware badge shows update-available if behind
AI Yield Insight Card	Season prediction vs target + irrigation deficit recommendation	Accept / View Evidence / Defer inline actions without leaving dashboard

Element	Description	State / Behaviour
Bottom Tab Navigation	5 tabs: Home, Crops, Devices, Trace, Search	Active tab in blue; all tabs have icon + label; 44px tap targets
Offline Banner	Amber banner when offline: "Last updated X minutes ago"	Most recent IoT data cached; CCR drafts queued for reconnection



Design Notes

Financial workbench uses identical component patterns to software SDE (SC-06) but with domain-adapted labels and content. Performance metrics use FONT_MONO for numeric values to ensure precise readability. Risk thresholds show contextual ✓/✗ indicators next to each metric value.

Element Annotations

Element	Description	State / Behaviour
Workbench Header	"Research Workbench" label — financial domain vocabulary adaptation	Sidebar reads "Research WBs" not "SDEs"; vocabulary layer active via PersonaContext
Primary Actions	"Open Jupyter" (primary), "Run Backtest", "Paper Trade", "Take Snapshot"	Paper Trade = pre-production deployment action for algorithms
Strategy Environment	Strategy name, universe, lookback, rebalance schedule, training dataset	Training data shows version + checksum link — audit of data lineage
Performance Dashboard	Sharpe Ratio, Max Drawdown, Annual Return, VaR with threshold indicators	Green ✓ = within risk limit; Red ✗ = limit breach; metrics update on each backtest
Risk Gate Status	4 gates: Sharpe, Drawdown, VaR, Liquidity — each with evidence link	Liquidity gate pending = strategy cannot proceed to live; blocks release

Element	Description	State / Behaviour
Drift Monitor	Model stability status over last 30 days with last-alert timestamp	AI drift detection — triggers recommendation when drift exceeds threshold
Regulatory Status	MiFID II, MAR, pre-trade controls compliance indicators	Domain-Pack-driven section; auto-checks on each backtest run

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2,847 artefacts

Upload Artefact

Filter

Artefact Name	Type	Version	SHA-256	Created	Status	
formulation-a-spec-v2.1.pdf	Specification	v2.1.0	a3f1c7d8...	3h ago	Verified	↓ ✓ ↗
stability-test-report-sim031.pdf	Test Report	v1.0	b8e2d1f4...	1d ago	Verified	↓ ✓ ↗
batch-record-B221.pdf	Batch Record	v1.0	c9f3e2g5...	2d ago	Verified	↓ ✓ ↗
ccr-042-approval-evidence.pdf	Compliance	v1.0	d1a4b7h6...	3d ago	Verified	↓ ✓ ↗
formulation-a-spec-v2.0.pdf	Specification	v2.0.0	e5c8d2i7...	14d ago	Archived	↓ ✓ ↗
lims-import-20250307.json	LIMS Data	v1.0	f6d9e3j8...	3h ago	Verified	↓ ✓ ↗
ingredient-subst-simulation.pdf	Test Report	v1.0	g7e1f4k9...	2d ago	Verified	↓ ✓ ↗

Design Notes

The artefact registry uses the same table pattern as the solution list (SC-04). SHA-256 digests use FONT_MONO in T_MUTED colour. The "Verified" state chip uses GREEN; "Archived" uses T_MUTED. Download, verify, and provenance actions use icon + text shorthand (↓ ✓ ↗) for compact display.

Element Annotations

Element	Description	State / Behaviour
Registry Header	Title + total artefact count chip + Upload Artefact (primary) + Filter	Upload triggers drag-and-drop zone; artefacts are immutable on upload
Artefact Name Column	File icon + name — icon adapts to artefact type (PDF, JSON, container)	Click name opens artefact detail: full SHA-256, provenance, download history
Type Badge	Specification, Test Report, Batch Record, Compliance, LIMS Data	Domain-specific types added via Domain Pack configuration
SHA-256 Digest	Truncated (first 8 chars) with full value on hover in tooltip	Integrity verification re-hashes and compares on demand; mismatch shows red alert

Element	Description	State / Behaviour
Verified Status	Green "Verified" chip after integrity check; "Archived" for superseded versions	Integrity failure shows red "Compromised" badge + immediate security alert
Download / Verify / Provenance	Three inline actions per row	Download logs who downloaded and when; Provenance opens full chain-of-custody
Artefact Immutability	All uploaded artefacts are write-once; no overwrite or delete permitted	New version creates a new record; prior versions remain accessible and auditable

QALAU

Universal OS

Acme Pharma Factory

🔍 Search or run command...

📱

+

JK

Account

Persona

Notifications

API Keys

Integrations

Security

Appearance

Help

Account Settings

Profile

EV

Change Avatar

Full Name

Elena Vasquez

Email

e.vasquez@acmepharma.com

Role

Formulation Lead

Organisation

Acme Consumer Goods

Notification Preferences

CCR approvals

On

Security alerts

On

AI recommendations

On

Build failures

On

Daily digest

Off

Persona & Vocabulary

Active Persona:

CPG / Pharma Chemist

This persona adapts the UI vocabulary to your domain.

• 'SDE' shows as 'Formulation Workbench'

• 'Build' shows as 'Simulation Run'

• 'Release' shows as 'Batch Release'

Change Persona

Security

Last login:

Today 08:45 · Dublin, IE

MFA:

Enabled (TOTP)

Sessions:

2 active sessions

Manage Sessions

Change Password

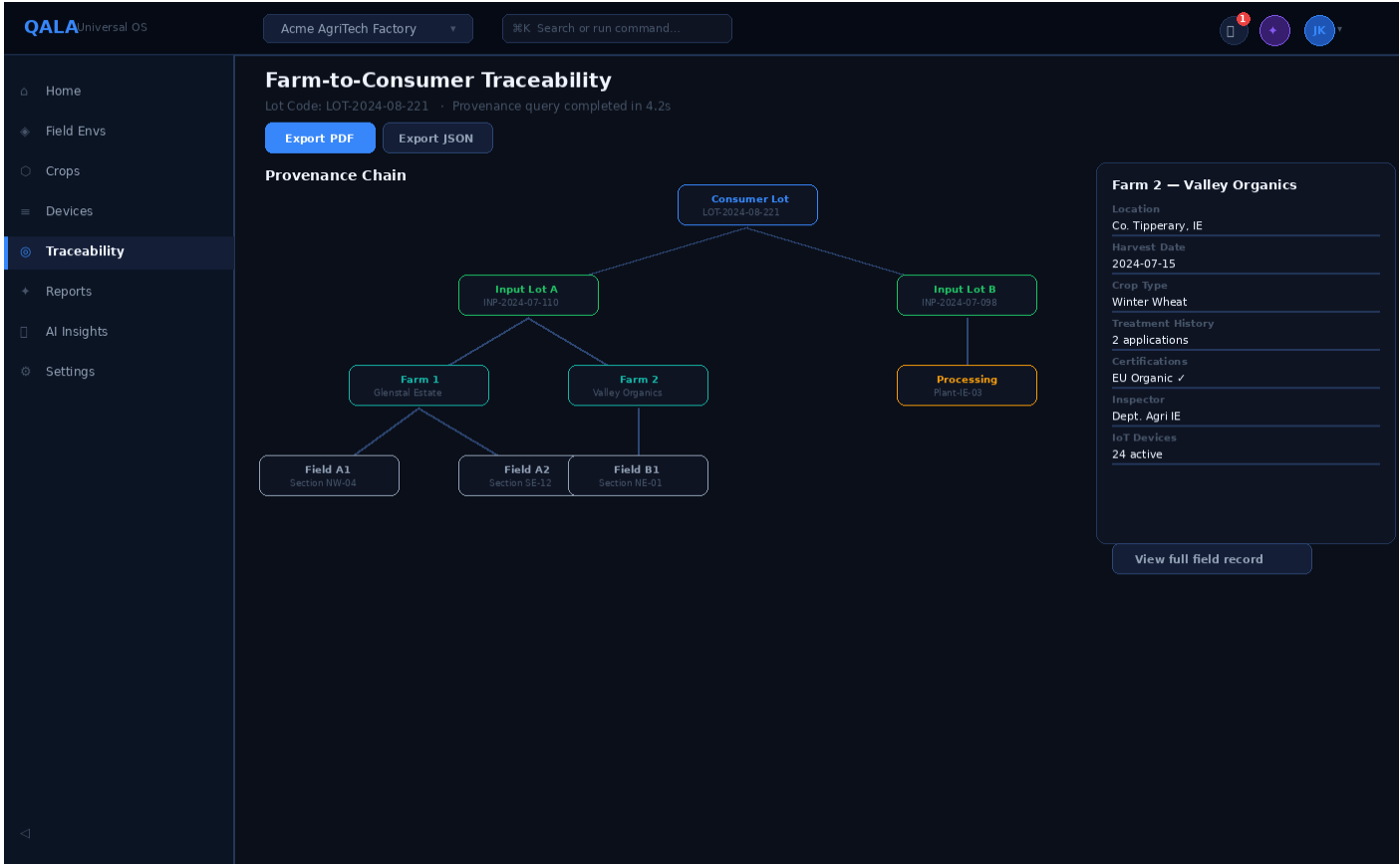
Design Notes

Settings use a left-sidebar + two-column-card layout. All form inputs use the same TextInput style as CCR forms. The avatar circle uses a 80px diameter with the user's initials as fallback. Security-related settings have amber "Requires MFA confirmation" indicators.

Element Annotations

Element	Description	State / Behaviour
Settings Sidebar	Account, Persona, Notifications, API Keys, Integrations, Security, Appearance, Help	Settings sidebar is independent of the main app sidebar
Profile Card	Avatar + editable profile fields: name, email, role, organisation	"Change Avatar" opens crop-and-upload modal; all changes require MFA confirmation
Persona & Vocabulary Card	Active persona chip + vocabulary impact list	"Change Persona" opens the same persona selector as SC-02 Onboarding
Notification Preferences	Per-category toggles: CCR approvals, security alerts, AI recommendations, builds, digest	Push, email, and in-app channels configurable per category

Element	Description	State / Behaviour
Security Card	Last login with location, MFA status, active session count	"Manage Sessions" lists all active sessions with revoke controls per device
MFA Status	"Enabled (TOTP)" shown in green	FIDO2 hardware key option available for compliance-tier tenants
API Keys	Not shown — accessible via sidebar; allows generating personal API tokens	Keys are scoped and expirable; activity log per key



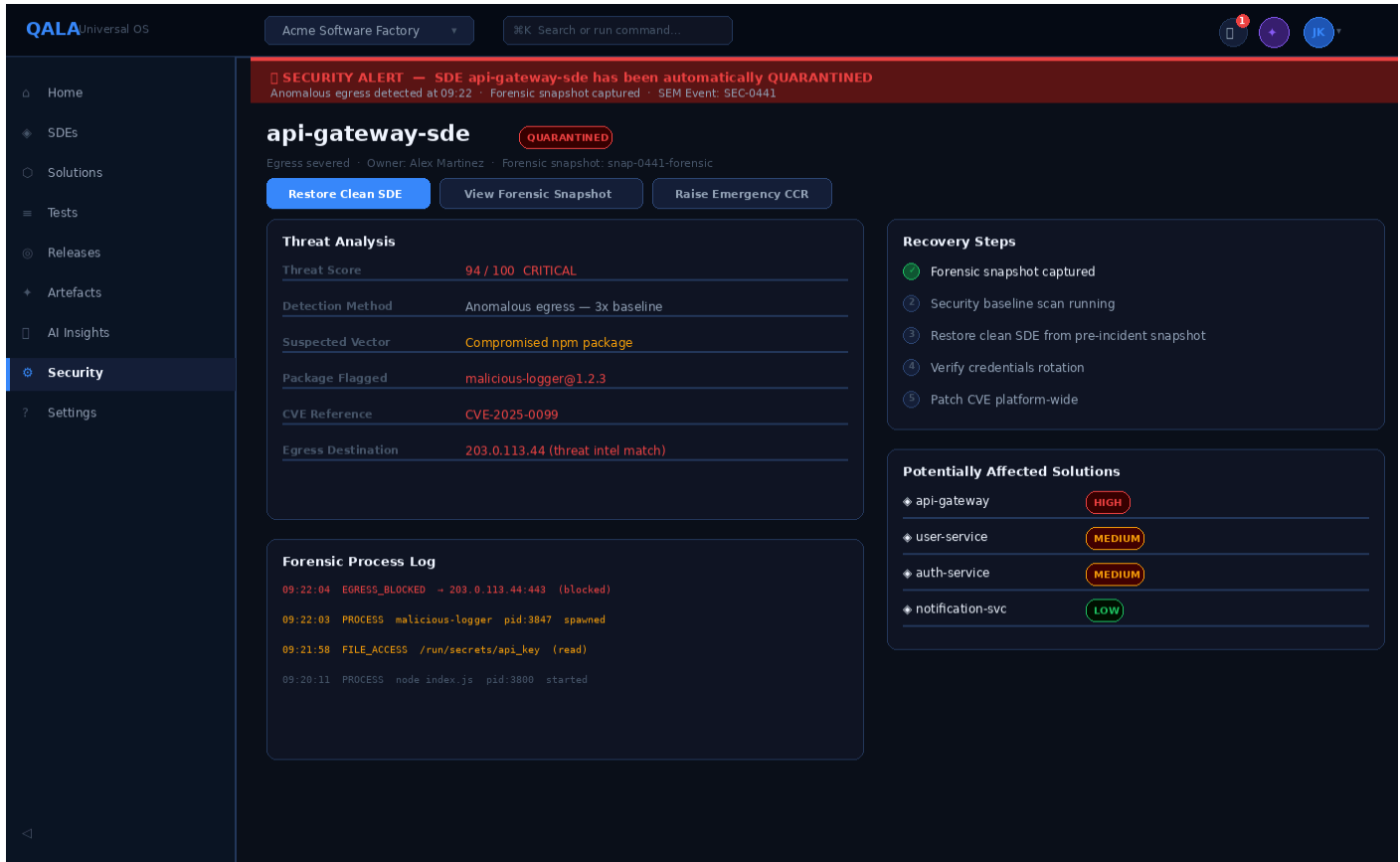
Design Notes

The provenance tree uses radial layout with nodes centred on their vertical positions. Node colours use the solution/domain accent colours to visually group by tier. The detail panel on the right is a 300px wide card that replaces generic drawer patterns for this screen.

Element Annotations

Element	Description	State / Behaviour
Query Result Header	Lot code, query completion time, export actions	Provenance query executes in < 10 seconds across distribution + batch records
Provenance Tree	Interactive tree: Consumer Lot → Input Lots → Farms → Fields	Nodes are clickable to expand detail; zoom + pan on larger screens
Consumer Lot Node	Top-level node: consumer product lot code	Clicking shows distribution date, retailer, and cold chain events
Input Lot Nodes	Mid-tier nodes linking consumer product to raw material lots	Colour coded by origin farm certification status
Farm Nodes	Farm name + location; clicking opens farm detail panel on right	Shows IoT fleet size, certifications, and inspector details

Element	Description	State / Behaviour
Field Nodes	Lowest-level provenance: field section, harvest date, treatment history	Agronomic records accessible; IoT sensor history linked
Detail Panel	Selected Farm 2 — Valley Organics with 7 detail fields	All fields from the GS1 EPCIS event record; PDF export includes this panel



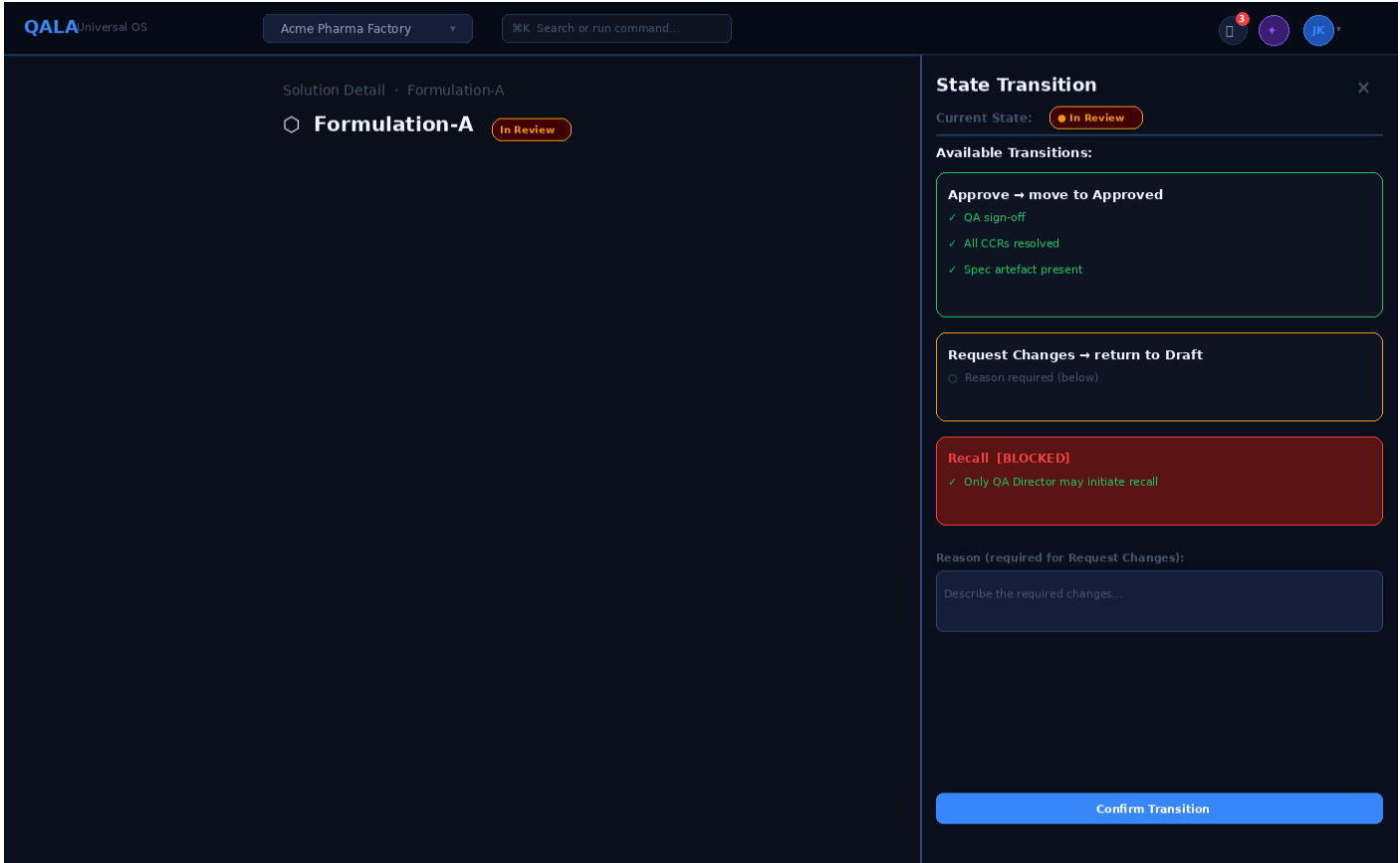
Design Notes

The security incident screen uses RED throughout to communicate urgency: alert banner, state chip, threat score badge. The forensic log uses the same near-black terminal background as the pipeline log (SC-07). The recovery steps panel uses the same vertical timeline as the approval chain (SC-08).

Element Annotations

Element	Description	State / Behaviour
Alert Banner	Full-width red alert banner: "SECURITY ALERT — SDE api-gateway-sde QUARANTINED"	Appears across all screens simultaneously for the affected SDE owner + security team
Quarantined State Chip	"QUARANTINED" chip with red pulsing dot animation	Cannot be dismissed without completing the recovery workflow
Action Bar	Restore Clean SDE (primary), View Forensic Snapshot, Raise Emergency CCR	"Restore Clean SDE" triggers a 5-step recovery workflow with confirmation dialog
Threat Analysis Panel	Threat Score 94/100 CRITICAL + detection method + suspected vector + CVE reference	All values are from SEM event payload; CVE reference links to NVD database
Forensic Process Log	Chronological process tree: egress blocked, malicious process spawned, file access	Timestamps in milliseconds; process IDs + file paths recorded immutably

Element	Description	State / Behaviour
Recovery Steps Panel	Numbered steps with status: done (✓ green), running (🔄 blue), pending (⏸ grey)	Step 1 auto-completes on quarantine; steps 2-5 require human actions
Affected Solutions Panel	Solutions potentially affected by the compromised package — sorted by severity	HIGH = immediate patch required; creates emergency CCR for each



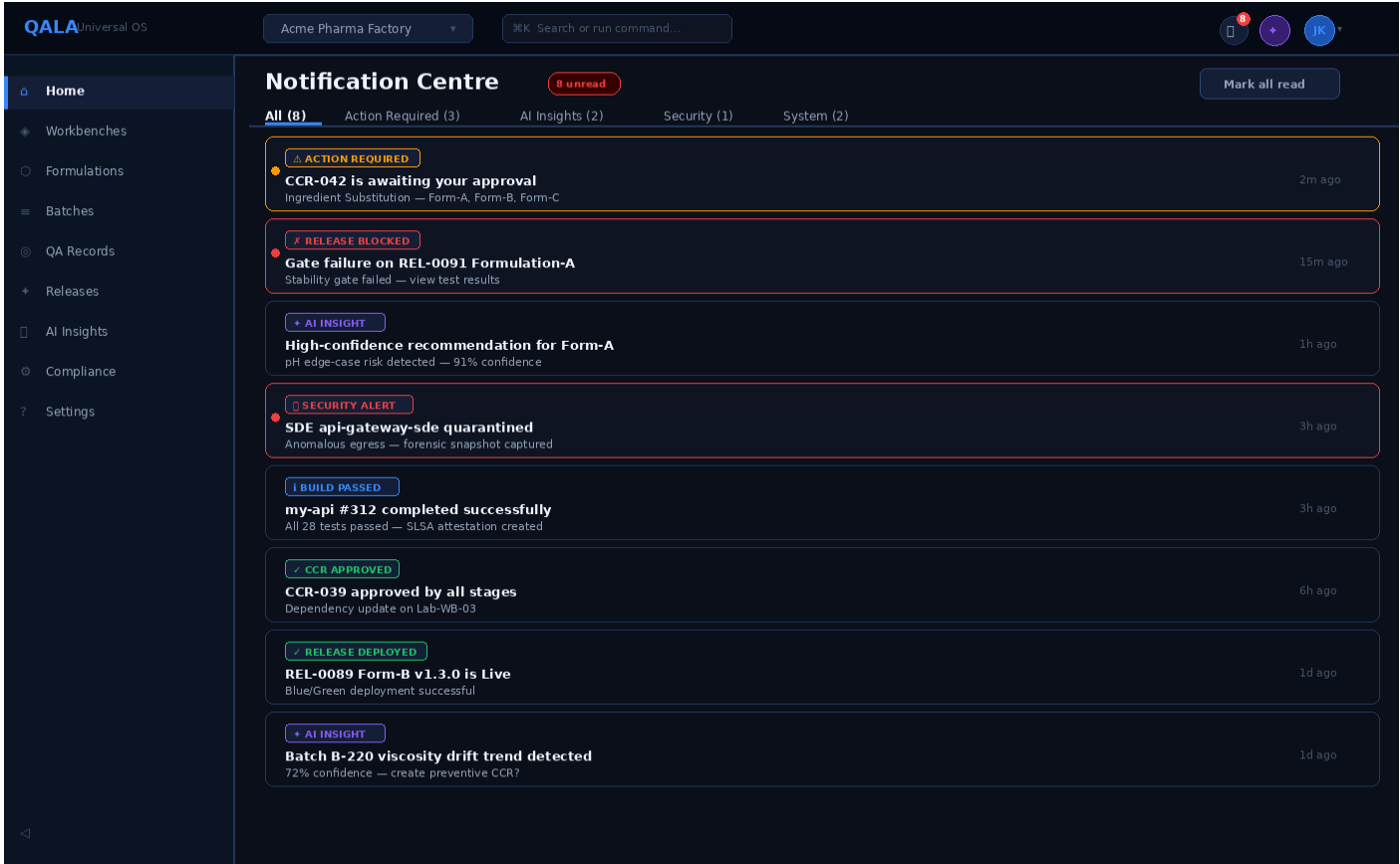
Design Notes

The state transition panel is a 480px right-side drawer. Each transition card uses a distinct colour (GREEN/AMBER/RED) for its border and accent. Blocked transitions use RED_DIM fill to visually differentiate from interactive options. The entire panel pattern is reused unchanged across all solution types and domains.

Element Annotations

Element	Description	State / Behaviour
Current State Display	"In Review" shown prominently at top with amber chip	State chip is the same component as used inline in all solution headers
Available Transitions	Cards showing: Approve → Approved, Request Changes → Draft, Recall [BLOCKED]	Each transition shows all preconditions with ✓ (met) or ○ (unmet) indicators
Approve Transition	All 3 preconditions met (✓) — QA sign-off, CCRs resolved, spec artefact present	Confirm button activates only when a valid transition is selected
Request Changes	Requires a reason text field — Confirm button disabled until reason entered	Minimum 20 characters required; reason stored with audit event
Recall [BLOCKED]	Red-accented card showing "Only QA Director may initiate recall"	Blocked transitions are visible but cannot be selected; show why they are blocked

Element	Description	State / Behaviour
Precondition Checklist	Inline ✓/○/🛑 per precondition in each transition card	Unmet preconditions link to the blocker (e.g. "1 critical defect must be resolved")
Confirm Button	Full-width primary button at bottom of drawer	Disabled until valid transition selected; triggers transition + AUDIT_EVENT publish



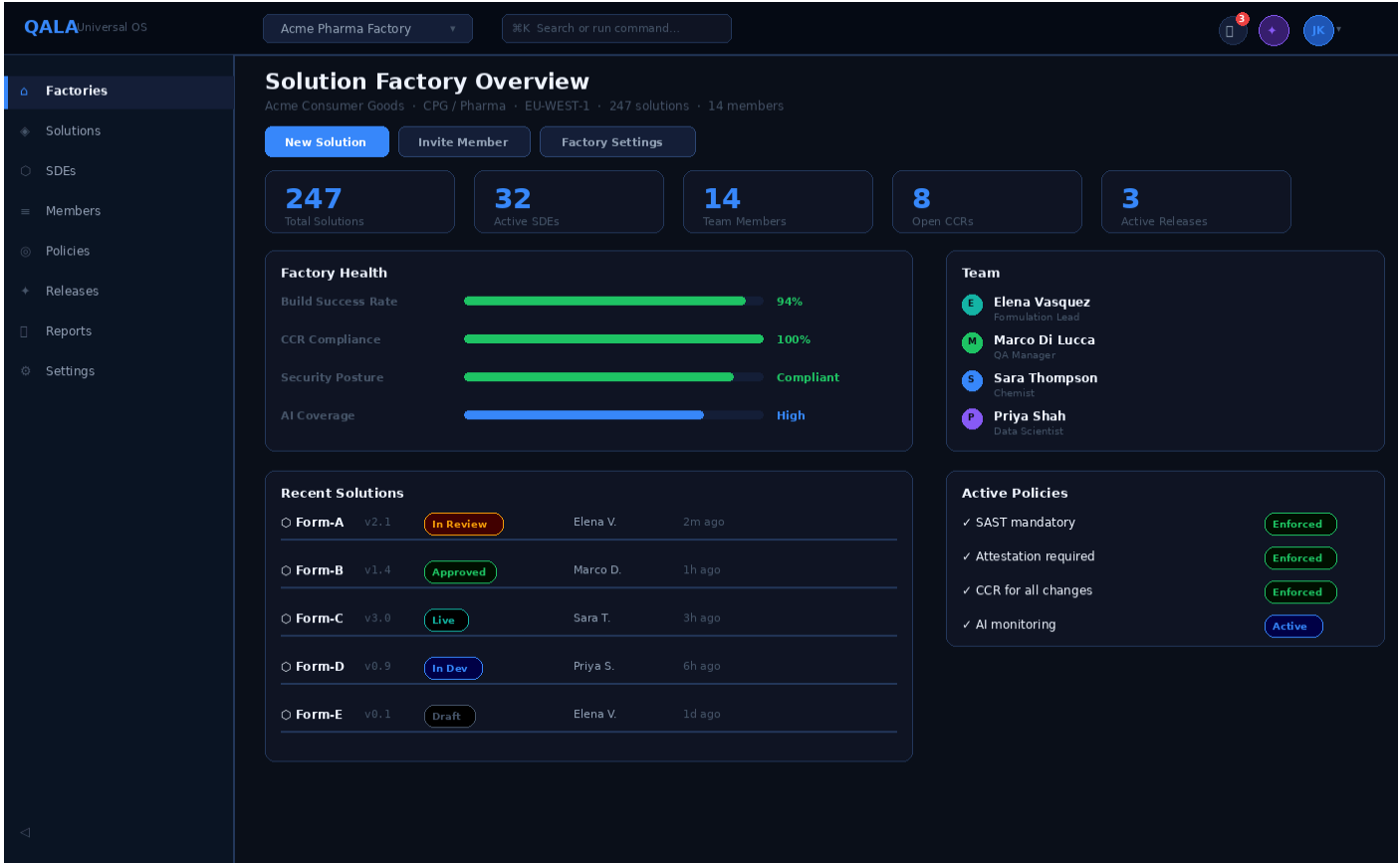
Design Notes

Notification cards use the left-border colour pattern consistently with all other status indicators across the platform. The unread dot sits outside the card at the left edge. Timestamps use relative format with absolute on hover. Error/security notifications are persistent until explicitly dismissed.

Element Annotations

Element	Description	State / Behaviour
Filter Tabs	All, Action Required, AI Insights, Security, System — with counts	Active tab has blue underline; counts update in real time
Mark All Read	Top-right action button — clears all unread indicators	Individual notifications can be dismissed with × on hover
Unread Indicator	Small dot on the left side of each unread notification card	Dot colour matches notification type (red = security, amber = action, purple = AI)
ACTION REQUIRED Card	Amber left border + bold badge + title + detail + timestamp	Direct action link within card; "Review Now" routes to CCR list filtered to user

Element	Description	State / Behaviour
RELEASE BLOCKED Card	Red left border — release is gated until action is taken	Click routes directly to the Release Pipeline screen (SC-09) for the blocked release
AI INSIGHT Card	Purple border — recommendation ready for review	Click opens the AI Agent Panel (SC-13) scoped to the specific recommendation
SECURITY ALERT Card	Red border + bold badge — highest urgency level	Click routes to SC-19 Security Incident screen for the quarantined SDE



Design Notes

The factory overview is the admin entry point for a Solution Factory. The 5-stat metric row uses large 28pt numbers in BLUE to give at-a-glance health status. Health bars show the overall operational posture. Policies are read-only indicators for non-admins; admins can click to toggle.

Element Annotations

Element	Description	State / Behaviour
Factory Summary Bar	Factory name, domain type, region, solution count, member count	Edit Factory Settings accessible to Factory Admin role only
Stats Row	5 metric cards: Total Solutions, Active SDEs, Team Members, Open CCRs, Active Releases	All counts are live; click any card filters the factory view to that resource type
Factory Health Bar	Build Success Rate, CCR Compliance, Security Posture, AI Coverage with progress bars	All bars update in real time; below-threshold bars turn amber/red
Recent Solutions	Latest 5 solutions across all types with state, owner, timestamp	Click any row goes to that solution's detail (SC-05)
Team Panel	4 members shown with role labels and avatar initials	"Invite Member" opens email invitation modal with role assignment

Element	Description	State / Behaviour
Active Policies Panel	Factory-level enforcement policies: SAST mandatory, attestation required, CCR for all changes, AI monitoring	Policy chips show "Enforced" (green) or "Off" (grey); policies set by Factory Admin
Factory Actions	"New Solution" (primary), "Invite Member", "Factory Settings"	New Solution opens the solution creation wizard scoped to this factory

4. Accessibility & Interaction Summary

All 22 screens comply with WCAG 2.1 Level AA. Key implementation requirements:

Element	Description	State / Behaviour
Contrast	All text meets 4.5:1 minimum; primary text achieves 7:1 AAA	Never use colour as sole status indicator — always pair with icon + text label
Focus Rings	3px blue (#388BFD) outline, 2px offset on all interactive elements	Never suppress :focus-visible; custom ring replaces browser default universally
Keyboard Nav	All interactive elements reachable via Tab; arrow keys within groups	G+H = Home, G+S = Solutions, G+E = Environments, Cmd+K = Command Palette
Screen Readers	All status chips have aria-label; live regions on AI + pipeline updates	All icon-only buttons have aria-label; decorative icons are aria-hidden
Motion	prefers-reduced-motion respected; all animations replaced with instant/opacity	AI orb, pipeline stage ring, toast slide-in all affected by this setting
Touch Targets	44x44px minimum for all interactive elements; 8px gap between targets	Bottom tab nav uses 44px height with 8px vertical padding
Mobile Zoom	All screens functional at 200% text zoom without horizontal scroll	Sidebar collapses to icon-only at md breakpoint; drawer at sm/xs

5. Component Inventory

The following design components appear across the 22 screens. Each is implemented as a reusable React component in the `@qala/design-system` package.

Element	Description	State / Behaviour
GlobalNav	Screens: all authenticated	Factory switcher + Command Palette trigger + notifications + AI orb + avatar
Sidebar Navigation	Screens: all except Login + Mobile	Domain-adaptive labels; 240px expanded / 64px icon-only collapsed
LifecycleState Chip	Screens: SC-03–SC-22	Dot + label; 14 distinct states; WCAG AA on all background combinations
SolutionCard	Screens: SC-03, SC-04, SC-05	Used in grid and list views; skeleton shimmer on load
QualityGateProgress	Screens: SC-07, SC-09	Horizontal pipeline node pattern; ✓ / ⚠ / 🛑 / ✗ states
AIRecommendationCard	Screens: SC-03, SC-05, SC-13	Purple accent; confidence bar; Accept/Dismiss/Defer actions
CommandPalette	Screen: SC-12 (accessible from all)	Cmd+K; 600px wide; grouped results; < 100ms response
EventStreamFeed	Screens: SC-03, SC-10	Real-time log with slide-in animation for new events
CCRFForm	Screen: SC-08 (creation mode)	Multi-section adaptive form; live risk score; cascade preview
StateTransitionPanel	Screen: SC-20 (right drawer)	Precondition checklist; typed justification for high-risk transitions
ArtefactCard	Screen: SC-16	SHA-256 digest; integrity verify; provenance chain link
ConfirmationDialog	Used by: state transitions, SDE decommission, recall initiation	Severity-coded; typed solution name required for critical actions